



Bharati Vidyapeeth's College of Engineering

A-4, Paschim Vihar, New Rohtak Road, New Delhi-63

**Applied Physics-I
Paper Code: ETPH-151**

Lab Manual

Department of Applied Sciences

Instructions for Writing the Experiments

1. All the experiment (left hand page of book of note book) and experiment number should be mentioned.
2. Following order must be maintained in writing the practical: Aim, Apparatus, Theory & Formula, Observation, Calculation, Result, Percentage error, Precautions and sources of error.
3. Procedure should not be written in any experiment.
4. Calculation should be shown on the left-hand side with pen.
5. The experiments should be supported by proper graphs where needed. Proper labeling of graph and scaling has to be done.
6. Percentage error should be calculated in all the experiments.
7. The observation in practical file should match the observation taken on observation notebook that has been signed by faculty.
8. Do all the calculation in the lab, including the error analysis.

Bibliography

Here is a short list of references to books which may be useful for further reading in Physics or instrumentation relevant to the experiments. Also included are some references to books of general interest with regard to science and experimentation.

1. "Fundamentals of Physics", 6th Ed., D. Halliday, R. Resnick and J. Walker, John Wiley and Sons, Inc., New York, 2001.
2. "The Feynman Lectures in Physics (Vols. 1, 11 and 111)", R.P. Feynman, R.B. Leighton and M.Sands, Addison Wesley, 1963.
3. "Students Reference Manual for Electronic Instrumentation Laboratories", S.E. Wolf and R.F.M. Smith, PHI, 1990.
4. "Laboratory Experiments in College Physics", C.H. Bernard and C.D. Epp, John Wiley and Sons, Inc., New York, 1995.
5. "Practical Physics", G.L. Squires, Cambridge University Press, Cambridge, 1985.
6. "Great Experiments in Physics", M.H. Shamos, Holt, Rinehart and Winston Inc., 1959.
7. "Experiments in Modern Physics", A.C. Melissinos, Academic Press, N.Y., 1966.
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9. "Practical Physics" Indu Prakash.

LIST OF EXPERIMENTS

1. To determine the wavelength of sodium light by Newton's Ring method.
2. To find the wavelength of sodium light by Fresnel's Biprism.
3. To determine the wavelength of spectral lines using diffraction grating.
4. To determine the refractive index of a prism using spectrometer.
5. To determine dispersive power of a prism using spectrometer & mercury source.
6. To determine the specific rotations of cane sugar solution with the help of half shade polarimeter.
7. To determine the wavelength of He-Ne LASER using transmission grating.
8. To determine the Numerical aperture of an Optical fiber.
9. To plot the a graph between distance of the knife edge from the centre of gravity and time period of bar pendulum. From the graph, find (a) the acceleration due to gravity (b) the radius of gyration.
10. To determine the Planck's Constant.

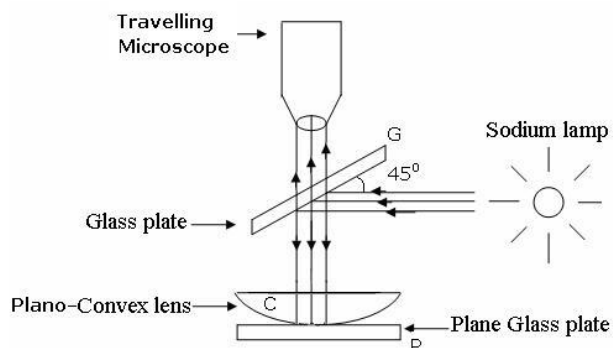
Experiment 1

Objective: To determine the wavelength of sodium light by Newton's rings.

Apparatus: Newton's rings apparatus (i.e, a planoconvex lens and an optically plane glass plate placed-on a wooden stand having a plane glass plate), a travelling microscope, sodium lamp, a convex lens and a spherometer.

Experimental set-up and theory:

Students are advised to go through the chapter 1 on interference, particularly article 1.11. The experimental set up for Newton's rings experiment is shown in Fig: 1.1. In it, a planoconvex lens (L) large focal length is placed with its convex surface in contact with an optically plane glass plate (P). An air film is thus formed between the lower surface of the lens (L) and the upper face of the glass plate (P). Monochromatic light from an extended source (S), (sodium lamp) placed at the focus of a convex lens (L) is allowed to fall on a glass plate (G) held at an angle of 45° with the vertical. After reflection from plate (G) light falls almost normally on the air-film.



A large number of concentric alternate bright and dark rings are formed (due to interference) in the air-film, which can be seen in a magnified form through a low-power microscope focussed on the film.

Theory:- Newton's rings are formed as a result of interference between the waves reflected from the top and bottom of air-film surface formed between the lower convex surface of the lens and the top surface of the plate P.

In Fig. 1.2, corresponding to an incident ray AB, the interfering rays are BD and CE and the rings are observed in reflected light. The formula used to calculate the wavelength of sodium light is given :-

$$\lambda = \frac{D_{n+m}^2 - D_n^2}{4mR}$$

Where,

D_{n+m} is the diameter of $(n+m)^{\text{th}}$ bright ring.
 D_n is the diameter of n^{th} bright ring.
 R is Radius of curvature of the curved surface of plano-convex lens.

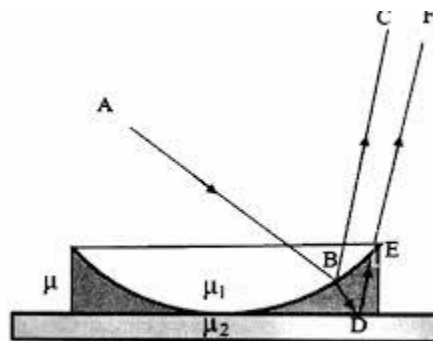


Fig. 1.2 Interference due to Air

The radius of curvature R is calculated with the help of spherometer using the formula as

$$R = \frac{l^2}{6h} + \frac{h}{2}$$

where l is the distance between the outer legs of the spherometer and h is the elevation of the tip of the central leg as compared to the outer legs when placed on the convex surface of the plano convex lens.

Procedure:-

(A) Determination of radii of bright rings: ' D_n '.

1. Clean the plano convex lens and the two glass plates and fix them as shown in Fig. 1.1
2. Set the glass plate G at an angle of 45° .
3. Switch on the sodium lamp and adjust the position of convex lens L_1 , so that a parallel beam of light falls on the inclined plate.
4. Place the travelling microscope with its tube vertical, above the inclined plate G with its objective above the point of contact of plano-convex lens L and the plate P .
5. Focus the eye-piece on the cross-wires and focus the microscope so that Newton's rings are clearly visible.
6. Calculate the least count of the microscope scale.
7. Clamp the microscope and using its slow motion screw move the microscope horizontally, till the cross-wire lies beyond 20^{th} or 24^{th} bright ring on the right hand side of the central dark ring.
8. Note microscope (cross-wire) reading on 20^{th} or 24^{th} bright ring.
9. Now slowly move the microscope towards left and note at least 8-10 readings, alternatively on bright rings e.g. on 24, 22d, 20th 2nd bright rings.
10. Now go on moving the microscope slowly towards left till it crosses the central dark point and again touches 2^{nd} bright ring.
11. Again note the readings on 2nd, 4h, 20th, 22d, 24h bright rings.
12. Here, care must be taken to note, only those bright rings on left hand side, corresponding to which reading on right hand side were noted.
13. Calculate the diameter D_n and D_n^2 of each n^{th} bright ring.
14. Plot a graph between D_n^2 , and n , which is a straight line (Fig. 1.4) and then find its slope.

(B) Determination of radius of curvature R

15. Remove the plano-convex lens from the arrangement.
16. Now take a spherometer and determine its pitch and least count.
17. Place the spherometer, first on the curved surface of the plano convex lens and after adjusting its central-screw, so as to touch the curved surface, note the reading.
18. Now place the spherometer on a plane glass plate P and again adjust the central screw so as to touch the plane surface and note the reading.
19. Calculate the value of h , Fig. 1.3(a).
20. Find the value of h at least three or four times and then take mean value of h .

21. To find the value of l , we take the impression of the tips of the legs (of the spherometer) on a note book and using the two upper tips of the jaws of vernier callipers, note the distance/between the three legs. Fig. 1.3(b).
22. Now take the mean value of l .
23. Calculate the value of radius of curvature R , with the help of the formulas.
24. Calculate the value of λ

Observations

Measurement of diameter of rings, D_n and D_n^2 .
Vernier constant reading of microscope.....cm.

S. No.	Number of bright rings	Microscope reading with cross-wires		Diameter of the ring $x - y = D_n$	D_n^2 (cm ²)
		On Right x (cm)	On Left x (cm)		
	24				
2	22				
3	20				
4	18				
5	16				
6	14				
7	12				
8	10				
9	8				
10	6				

(ii) Measurement of radius of curvature R .

Pitch of the spherometer =.....cm

Total no. of divisions on circular scale =.....cm

Least count of spherometer =.....cm

S. No.	Reading of Circular Scale		No. of complete rotation of circular scale in moving from convex to plane surface (n)	$h = n \times \text{pitch} + (a-b) \times \text{LC}$ or $h = n \times \text{pitch} + (100 + a - b) \times \text{LC}$ (cm)
	on convex surface (a)	on plane surface (b)		
1				
2				
3				

The distance between the legs of spherometer (Fig 1.3 (b))

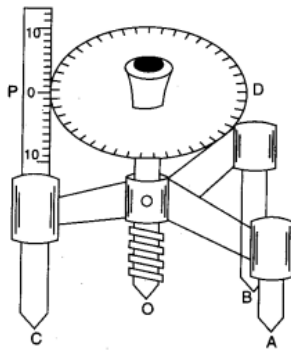


Fig. 2.14. Spherometer.

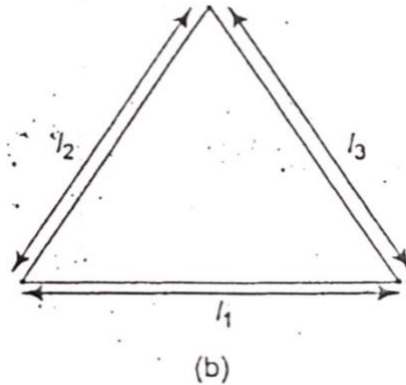


Fig. 1.3 (a) Measurement of h

(b) measurement of L

$L_1 = \dots\dots\dots \text{cm};$ $L_2 = \dots\dots\dots \text{cm};$ $L_3 = \dots\dots\dots \text{cm}$

Mean value of $L = \frac{L_1 + L_2 + L_3}{3} = \dots\dots\dots \text{cm}$

Calculations: Graph is plotted between D_n^2 and n as shown in Fig. 1.4.

The radius of curvature of the convex surface of the plano-convex lens is

$$R = \frac{l^2}{6h} = \dots\dots\dots \text{cm}$$

2

For calculating the value of wave length by this method

$$\lambda = \frac{d_{n+m}^2 - D_n^2}{4mR} = \tan \theta = \dots\dots\dots \text{cm}$$

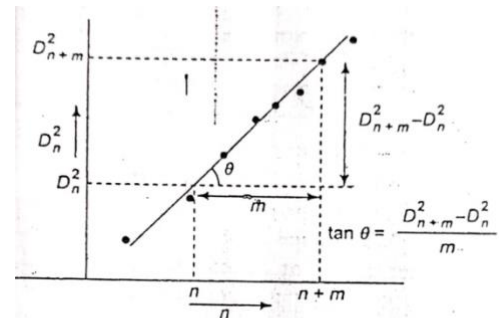


Fig. 1.4. Graph between D_n^2 and n

Knowing the value of $\tan \theta$ (i.e. slope of straight line between D and n) and R , the value of λ can be calculated.

Result: Wave length of sodium light as obtained from by Newton's rings method is $= \dots\dots\dots \text{cm}$.

Actual Value: The standard value of wave length of sodium light is $5.893 \times 10^5 \text{ cm}$.

Percentage Error: The percentage error is $\dots\dots\%$.

Precautions and Sources of Error

1. The surfaces of plano-convex lens and plane glass plate should be optically spherical and plane respectively.
2. The plano convex lens and the glass plate should be cleaned thoroughly.
3. The convex surface of the plano-convex lens should be large in aperture.
4. The sodium light should have an extended source.
5. Before starting the measurements for diameter of the rings, slow motion of the microscope only in one direction should be ensured for whole range of observations, only to avoid backlash error.
6. Cross-wire should be adjusted on the bright rings tangentially.
7. The value of 'h' in the measurement of radius of curvature should be measured accurately.
8. Newton's rings setup should not be disturbed during the whole set of observations.
9. The value of $(D_{n+m}^2 - D_n^2)/m$ should be calculated graphically to avoid errors involved.

VIVA-VOCE QUESTIONS

Q.1 How are Newton's rings obtained?

Ans. A convex lens is placed on a plane glass plate and when monochromatic light is allowed to fall on it vertically, it is reflected from the lower spherical surface of the lens and also from the upper surface of the plane glass plate. These two reflected rays on superposition produce interference of light and the rings are obtained.

Q.2 Why are Newton's rings circular?

Ans. The air film between the lens and the glass plate is of variable-thickness. Points of constant thickness in this film lie on circles with the point of contact of the lens and the glass plate at the centre. Since every fringe is the locus of a constant thickness of air film, it is circular.

Q.3 What should be the central spot brighter or dark?

Ans. In reflected light the central spot should be dark. The total path difference between the two interfering rays is $(2\mu t \cos \Theta = \lambda/2)$ where Θ is the angle of refraction in the film of thickness t . The path difference brought about by reflection from the glass plate is $\lambda/2$. At the point of contact $t=0$ and so the total path difference is $\lambda/2$ only. At the point of contact, the two interfering rays are opposite in phase and produce zero intensity. Hence the central spot is dark.

Q4. On what factors does the diameter of the rings depend ?

Ans. The diameter of the Newton's rings depends upon

- (i) the refractive index of the medium enclosed between the lens and the plate
- (ii) wavelength of light used
- (iii) the radius of curvature of the spherical surface of the lens.

Q.5. Will a plane mirror not serve the purpose better than a plane glass plate?

Ans. No rings will be visible with a plane mirror because no part of the light will be transmitted through it. Hence the reflected part and the transmitted part will superimpose over each other and uniform illumination will be produced.

Q.6. Where are the fringes formed?

Ans. The rings are formed in the air film between the glass plate and the lens. These are called local-ised fringes.

Q.7. Why do you use a convex lens in front of the sodium lamp?

Ans. The sodium lamp is kept at the focus of the convex lens so that a parallel beam of light falls on the inclined glass plate above the lens. If the lamp is kept a little away from the arrangement the convex lens need not be used.

Q.8. What will you get with a white light source instead of a sodium lamp?

Ans. We shall be able to see some coloured rings with a white light source. The diameters of the ring of different colours will be different and these will overlap on each other producing general illumination at most of the space. The central spot will be black for each wavelength and so it will be black for white light as well.

Q.9. What will be the change in the diameter of the rings if a drop of a transparent liquid is put between the lens and the plate?

Ans. The firings will contract because their diameter will become less, as given by the formula

$$D_n^2 = \frac{(2n+1)R}{2\mu\lambda} \text{ for a bright ring.}$$

Q.10. Why is an extended source used here?

Ans. So that whole surface of the lens is illuminated, otherwise the fringes will not be circular.

Q.11. Why should a lens of large radius of curvature be used?

Ans. Greater the radius of curvature of the lens, greater would be the diameters of the rings. This will reduce the percentage error in the experiment. Also with such lens the thickness of air film would be small as required by the theory.

Q.12. Are rings equispaced?

Ans. No with increase in diameter, the rings get closer and closer.

Q.13. Why do the rings get closer as we move away from the centre of the central ring?

Ans. From the formula, it is seen that the diameter of dark rings is proportional to the square root of natural numbers, while that of bright rings to the square root of the odd natural numbers. Hence, as the order of the rings increases the corresponding diameters do not increase in the same proportion. Due to this the rings get closer at an increasing rate.

Q.14. How do the rings appear when viewed in transmitted light?

Ans. The central ring will be bright.

Q.15: What will happen if the lens is placed on a plane mirror instead of a glass plate?

Ans. The rings will disappear. By reflection of the transmitted rays, the two complimentary systems of rings will superpose. On each other resulting in uniform illumination.

EXPERIMENT NO. 2

Objective: To determine the wavelength of sodium light using a Fresnel's biprism.

Apparatus: An optical bench four upright stands, a bi-prism, an adjustable eye-piece capable of reading upto 0.001 cm, a short focal length convex lens and a sodium lamp.

Theory: Before performing the experiment, students should read chapter 1, particularly the article 1.8. The experimental arrangement is shown in Fig. 2.1. The various devices are arranged at the same height above the bed of the optical bench.

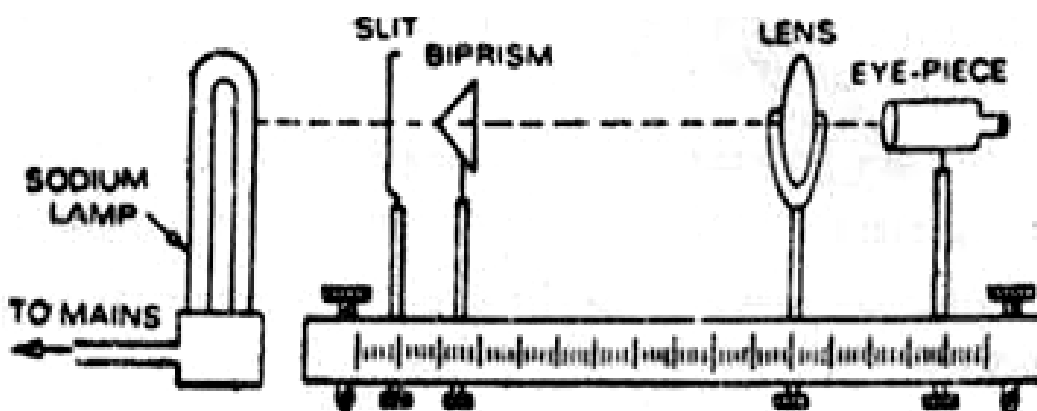


Fig. 2.1 Experimental Set-up

As discussed in chapter 1, a closely spaced interference fringes are produced by a bi-prism. The fringes are actually formed anywhere in the space between the biprism and the eye piece and are called the non-localized fringes. The fringes are observed with a microscope, with the fringes lying in the focal plane of its eye-piece.

If d is the distance between the sources S_1 and S_2 (Fig. 2.3), D is the distance between the slit S and the cross-wires of the eye-piece; and β is the fringe width then

$$\lambda = \frac{d}{D} \beta$$

Thus the wavelength λ of the monochromatic source can be determined by measuring D , d and β . To determine d , a convex lens is chosen such that its focal length is less than one-fourth of the distance between the slit and the focal plane of the eye piece. The lens is mounted on a stand which is kept between the stands holding the Fresnel's biprism and the eye-piece (Fig.2.3). The lens is so adjusted that for two of its positions the real images of the two virtual sources S_1 and S_2 are focused on the focal plane of the eye-piece. If d_1 and d_2 are the separation between the real images of S_1 and S_2 for two positions of the lens, the magnifications for the positions of the lens are

$$m_1 = \frac{d_1}{d} = \frac{v}{u} \quad \text{and} \quad m_2 = \frac{d_2}{d} = \frac{v_1}{u_1} = \frac{u}{v}$$

where u and v are the distances of the object and the image respectively from the lens in the first position L_1 (Fig. 2.3).

$$\frac{d_1}{d} \times \frac{d_2}{d} = \frac{v}{u} \times \frac{u}{v} = 1$$

$$d_1 d_2 = d^2$$

Hence,
$$d = \sqrt{d_1 d_2}$$

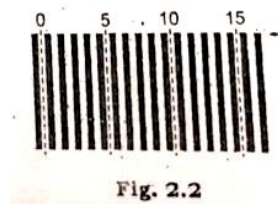
Procedure: - In this experiment, adjustment of the instrument are extremely important to carry out the experiment successfully.

The following adjustments should be done.

- (i) With the help of a spirit-level and the leveling Screws provided at the base of the instrument, level the bed of the optical bench.
- (ii) Mount the upright carrying the slit near the zero of the scale on the optical bench. Bring the slit in the middle of the bench by moving the slit-stand perpendicular to the bench. Make the slit very narrow by the screw for adjusting the slit-width and illuminate it by sodium light.
- (iii) Mount the biprism on its stand at about 15 cm from the slit such that the centre of the biprism and that of the slit are at the same height. See that the Slit and the edge between the faces of the biprism are vertical.
- (iv) Place the eye on the side of the biprism away from the slit. Two virtual images of the source will be seen. Move the eye across the bench, one of the images will appear to cross the edge of the biprism from one side to the other. If the refracting edge of the biprism is parallel to the slit, then the image as a whole will appear to cross the edge. If either the top or the bottom of the image crosses first, the adjustment is faulty. Adjust the biprism by rotating it in its own plane by the tangent screw provided for the purpose to cause the transition of the image as a whole. Now the slit, the cross-wire and the refracting edge of the biprism are exactly parallel.
- (v) Mount micrometer eye-piece on the optical bench about 80-90 cm from the slit. Adjust the eye-piece to get the fringes in its focal plane. If no fringes are visible, move the biprism upright at right angle to the bed of the optical bench till the fringes appear. Then make the fringes as distinct as possible by rotating the biprism in its own plane by the tangent screw.
- (vi) To remove the lateral shift, set the cross-wire on any bright fringe. If on moving the eye-piece away from the biprism, the fringe shifts away from the cross-wire, move the biprism perpendicular the optical bench till this shift is not there.

Measurements

1. Place the biprism upright near the slit upright and move the eye-piece away to a distance of above 80-90 cm from the slit. After doing all the necessary adjustments discussed above, set the vertical cross-wire of the eye-piece in the middle of a bright band



towards left (Fig. 2.2) and note the reading of the micrometer screw of the eye-piece. Next displace the eye-piece horizontally towards right till the cross wire comes in contact with the middle of the 5th bright band. Read the micrometer again. Then take the cross-wire to the 10th band, read the micrometer and so on. The difference of the two consecutive readings is the width of 5 fringes and the fringe width.

$$\beta = \frac{\text{width of 5 fringes}}{5}$$

2. Repeat step(1) for the horizontal movement of the eye-piece from right to left and find the mean fringe-width β .
3. Measure the distance D from the slit S to the cross wire of the eyepiece. The distance so measured on the scale may not be the correct distance and an error called the bench correction is introduced, which is explained in step 6.
4. To measure $S_1S_2=d$, place a convex lens of focal length $f < D/4$ in the position L_1 such that clear images of S_1 and S_2 are formed in the focal plane of the eye-piece (Fig. 2.3). Set the vertical cross-wire on images and note the micrometer reading. Move the micrometer screw till the vertical cross-wire is on the second image. Again note the micrometer reading. The difference between the two readings gives the distance d between the two images.

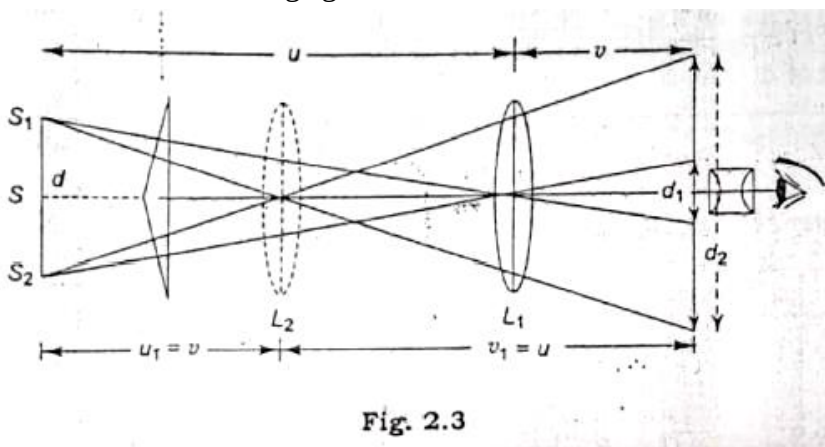


Fig. 2.3

5. Next displace the lens to the position L_2 such that clear images of S_1 and S_2 are again formed in the focal plane of the eye-piece. Measure their separation d . Then,

$$d = \sqrt{d_1 d_2}$$

6. To calculate the bench correction, remove the biprism and put a rod of measured length (l) between the slit and the focal plane of the eye-piece. Take the readings of the up rights of slit and the eyepiece. If l' is the observed distance between the uprights of the slit and the eyepiece, as read on the scale of the bench, then the quantity $(l-l')$ gives the bench error and is to be added to the measured value of D to get correct value of D .

Observations :-

Pitch of the screw =cm

No. of divisions on micrometer screw =cm

Least count of micrometer screw =cm

(i) Table for measurement of fringe width β .

S. No.	No. of band	Micrometer Reading	Width of 5 bands x	Fringe width $\beta = \frac{x}{5}$
Left→Right				
1	0	X_1		
2	5	X_2	$X_2 - X_1$	
3	10	X_3	$X_3 - X_2$	
4	15	X_4	$X_4 - X_3$	
5	20	X_5	$X_5 - X_4$	
6	25	X_6	$X_6 - X_5$	

Right→Left

Mean $\beta = \dots\dots\text{cm}$

Same observation as above

(ii) Measurement of d

S. No	Lens position L ₁		Distance between two images d ₁ (cm)	Lens position L ₂		Distance between two images d ₂ (cm)	d = √ d ₁ d ₂ (cm)
	Micrometer Reading			Micrometer Reading			
	1st Image	2nd Image		1st Image	2nd Image		
1							
2							
3							

Mean $d = \dots\dots\text{cm}$

Distance D between slit and eye-piece upright =cm

bench correction =cm

corrected $D = \dots\dots\text{cm}$

Calculations

$$\lambda = \frac{d}{D} \beta$$

=cm =m

Result: The wave-length of sodium light by Fresnel's biprism=.....m.

Actual value=.....

% Error=.....

Precautions and Sources of Error

1. The slit biprism and eye-piece must be at the same vertical height above the bench.
2. The slit should be made narrow. It should be made parallel to the vertical cross-wires of the eyepiece.
3. The bi-prism should be rotated in its own plane so that its edge becomes exactly parallel to the slit.
4. While taking measurements for the fringe width, the vertical cross-wire should be set in the centre of the bright fringe only.
5. For determining the distance d between the two virtual images of the slit, the lens should be so chosen that its focal length $f < \text{one-fourth}$ of the distance between the slit and the cross-wire.
6. If the eye piece happens to be too close to the biprism it may not be possible to obtain both images with the lens. The eye piece should be so placed that it is possible.
7. Bench correction should be measured and corrected distance between the slit and the cross-wire should be found.
8. While using the micrometer screw, back-lash error should be avoided. This can be done by rotating the screw in the same direction.

VIVA-VOCE QUESTIONS

Q.1. What is a biprism?

Ans. A biprism is a combination of two acute prisms placed base to base. This is made from an optically plane glass plate by proper grinding and polishing.

Q.2 Why are the refracting angles of the two prisms made so small?

Ans. By doing so the distance 'd' will be small and so the fringe width will be large.

Q.3 What is the purpose of biprism?

Ans. The biprism is used to produce two coherent images of a given slit which are separated at a distance and behave as two coherent sources.

Q.4 On what factors does the fringe-width depends?

Ans. The fringe width B is given by

$$\beta = \frac{D}{d}\lambda$$

- (i) The fringe width is directly proportional to the wavelength of light.
- (ii) It is directly proportional to D , the distance between source and screen. For small distance, the fringes will become dim, thus an optimum distance is a must.
- (iii) It is inversely proportional to d , the distance between two coherent sources. Attempts are made to keep 'd' small.

Q.5. How do you measure 'd'?

And. We use displacement method. In this method a convex lens is used and the formula used is

$$d = \sqrt{d_1 d_2}$$

Q.6. What will happen if a monochromatic source is replaced by white light source?

Ans. In case of white light, the interference pattern consists of a central white fringe surrounded by a few colored fringes.

Q.7. How will you locate zero order fringe in biprism experiment?

Ans. First the interference fringes are observed with monochromatic source of light. The monochromatic source is replaced by white light source. The cross wire is fixed at white fringe. White light source is replaced by monochromatic source. The fringe on cross wire is zero order fringe.

Q.8. Are the biprism fringes perfectly straight?

Ans. The biprism fringes are not straight but hyperbolic in nature. The eccentricity of the hyperbola is so large that they appear as straight in the field of view of eyepiece.

Q.9. Why do you get limited number of fringes in biprism experiment?

Ans. This is due to the inhomogeneity of the light source. The various components produce the own system of fringes of slightly different fringe-width. Due to their overlapping only a limited number of fringes are observed.

Q.10. Why should the slit be narrow?

Ans. The slit, is made narrow to obtain a good and distinct fringe pattern with contrasts. With wide slit, the fringes will be blurred with no contrast.

Q.11. Why is the distance between the slit and biprism made small while that between the slit and eyepiece large?

Ans. As the fringe width $\beta = \lambda D/d$, large D and small d helps in making fringes broad and reduces the error in the measurement of fringe-width.

Q.12. Where are the fringes formed?

Ans. The fringes are formed in space anywhere between the biprism and the eye-piece. They are not localized fringes.

Q.13. What happens when the angle of the biprism is doubled?

Ans. The fringes will become narrow, with their width halved.

Q.14. Will the fringe width remain the same for all colours of light?

Ans. No, as $\beta = \lambda D/d$, the width is directly proportional to the wavelength of the light used.

Q.15. What is the difference between biprism fringes and those obtained with Newton's rings?

Ans. (i) The biprism fringes are straight and equally spaced while Newton's rings fringes are circular and not equispaced (ii) Biprism fringes are obtained by division of wave front and in Newton Rings, fringes are obtained by division of amplitude.

EXPERIMENT NO. 3

Object: To determine the wavelength of sodium light using diffraction grating.

Apparatus: A plane diffraction grating, spectrometer, mercury lamp, prism, spirit level and magnifying lens.

Theory: Student should carefully read Art. 2.6 of chapter 2 on diffraction of light. The wavelength of any spectral line can be calculated by the formula.

$$(a+b)\sin\theta = n\lambda$$

$$\lambda = \frac{(a+b)\sin\theta}{n}$$

Where

$(a+b)$ = Grating element

θ = angle of diffraction

n = order of spectrum

Procedure: The whole procedure may be divided into three parts :-

(a) Adjustments of spectrometer: Following adjustments are made in the spectrometer:

1. The axis of the telescope and that of the collimator must intersect the principal vertical and rotation of telescope.
2. Prism table should be leveled.
3. Telescope and collimator are adjusted for parallel light.

(b) Adjustments of Grating: Following adjustment of the grating are made:

[A] To mount the grating with its plane vertical and to set it for normal incidence.

1. First set the telescope (T) to receive direct light from the collimator (C). Bring the telescope cross-wire at the direct image of the slit and take the reading of one vernier only.
2. Rotate the telescope through 90° and fix it there.
3. Place the grating (G) mounted in its holder on the prism table vertical and in such a way these:
 - i) The lines of grating are perpendicular to the table.
 - ii) The plane of the grating passes through that diameter of the prism table which is normal line joining the leveling screws P1 and P2 the table Fig. 3.1(a).
 - iii) The ruled surface extends equally on both sides of the centre of the prism table.
4. Turn the prism table until an image of the slit is formed on the cross wire of the telescope reflection from the unruled surface of the grating as shown Fig. 3.1(b).

5. Rotate one of the screws, P1 or P2 till the centre of the image coincides with the junction of cross wires. This makes the plane of the grating vertical. The light is now incident on the grating angle of 45° (Fig. 3.1b). Note the reading of the prism table.
6. Rotate the prism table through 45° in the proper direction so that the ruled surface of the grating faces the telescope. In this position, light from the collimator is incident normally on the unruled surface of the grating. Clamp the table in this position.

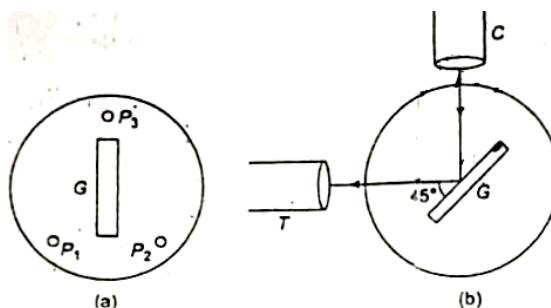


Fig. 3.1

[B] To make the rulings of the grating vertical.

The rulings of the grating may not be vertical even with its plane being so. In that case diffracted image is displayed upwards on one side of the direct image and downwards on the other side.

To make the rulings vertical turn the screws P3 of the prism table the image on both sides are symmetrical.

[C] Before starting the experiment, make the collimator slit very narrow so that the diffraction fringes are very sharp.

(c) The actual experimental procedure:-

The spectrum obtained in a grating is shown in Fig. 3.2.

- (i) Rotate the telescope to the left side of direct image and adjust the different spectral lines (violet, green and red) turn by turn on the vertical cross wire for 1st order. Note down the reading of both the verniers in each setting.
- (ii) Rotate the telescope further to obtain the second order spectrum and again the spectral lines on the vertical cross wire and note the reading.
- (iii) Now rotate the telescope to the right of the direct image and repeat the above procedure for first order as well as for second order.
- (iv) Find out the difference of the same kind of verniers (V_1 from V_1 and V_2 from V_2) for each spectral line in the first order and then in the second order. The angle is twice the angle of diffraction for that particular colour. Half of it will be angle of diffraction.
- (v) Find out the angles of diffraction for other colors in first and second orders.

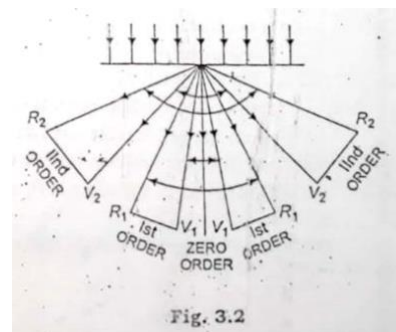


Fig. 3.2

Observation

Vernier Constant of the Spectrometer=

Number of lines on grating =15000/inch, width of grating: 2.54 cm

Least Count of Spectrometer =

$$\text{Grating element (a+b)} = \frac{2.54}{N} = \dots\dots\dots \text{cm}$$

Order of Spectrum	Vernier Scale Reading MSR + (VSR x LC)		Angular Width $2\theta_n$ (Degree)	Half Angular Width θ_n (Degree)	Average Angle θ_{av} (Degree)
	Telescope on LHS	Telescope on RHS			
I	V ₁ = V ₂ =	V ₁ = V ₂ =	2 θ_1 = 2 θ_2 =	θ_1 = θ_2 =	θ_{avI} =
II	V ₁ = V ₂ =	V ₁ = V ₂ =	2 θ_1 = 2 θ_2 =	θ_1 = θ_2 =	θ_{avII} =

Calculations :- The wavelength of various spectral lines can be calculated from

$$n\lambda = (a+b)\sin\theta_n$$

$$\lambda = \frac{(a+b)\sin\theta_n}{n}$$

For 1st order spectrum,

$$\begin{aligned} n &= 1 \\ \lambda &= (a+b) \sin\theta_1 \\ (a+b) &= \dots\dots\dots \text{cm} \\ \theta_1 &= \dots\dots\dots \text{cm} \\ \sin\theta_1 &= \dots\dots\dots \\ \lambda &= \dots\dots\dots \text{cm} = \dots\dots\dots \text{m} \\ \lambda &= \dots\dots\dots \text{m} \end{aligned}$$

Thus,

For 2nd order spectrum,

$$\begin{aligned} n &= 2 \\ \lambda &= \frac{(a+b)\sin\theta_2}{2} \\ (a+b) &= \dots\dots\dots \text{cm} \\ \theta_2 &= \dots\dots\dots \text{cm} \\ \sin\theta_2 &= \dots\dots\dots \\ \lambda &= \dots\dots\dots \text{cm} = \dots\dots\dots \text{m} \\ \lambda &= \dots\dots\dots \text{m} \end{aligned}$$

Thus,

Result: Mean value of $\lambda = \dots\dots\dots$

Precautions and Source of Error:

1. All adjustments of the spectrometer must be correctly done.
2. The rulings of the grating must be vertical.
3. The light must be incident on the unrulled side of the grating. Otherwise refraction occurs at the thick glass plate after diffraction and the angle measured will not correspond to that of diffraction alone.
4. The grating should be as narrow as possible and parallel to the rulings of the grating.
5. The slit should be narrow as possible and parallel to the ruling of the grating.
6. While taking reading of the telescope, the turn-table should be clamped and vice versa.
7. Readings should be taken in both the vernier to eliminate the error due to non-coincidence of the centre of the graduated scale with the axis of rotation of the spectrometer.
8. Light should be incident normally on the grating surface. Otherwise the angle of incidence would not be zero which is a condition for the derivation of the formula $(a+b)\sin\theta = n\lambda$. Further a small change in the angle of incidence cause a large error in the angle of diffraction.
9. Apart from the experimental errors, the grating itself may have the following errors:-
 - a) Error due to curvature and non-parallelism of the rulings.
 - b) The error due to variation of the rulings.
 - c) The error of spacing.

Viva-Voce Question:

Q1. What is diffraction grating?

Ans. An arrangement consisting of a large number of parallel slits of same width and separated by equal opaque spaces is known as diffraction grating.

Q2. What are the requisites of a good grating?

Ans. The lines should be exactly parallel, uniform, equidistant and of equal width.

Q3. What is grating element?

Ans. The distance between the centers of two successive slit is called grating element. This is denoted by $(a+b)$ where a is width of transparent part and b is width of opaque part.

Q4. How many orders do you get here? Why?

Ans. We know that $(n)_{\max} = \frac{(a+b)}{\lambda}$. The grating which we are using has 15,000 lines per inch.

Hence, $(a+b) = \frac{2.54}{15,000}$, $\lambda = 5893 \times 10^{-10} \text{m}$

$$(n)_{\max} = \frac{2.54}{15000 \times 5893 \times 10^{-10}} = 2.875 < 3$$

Thus, we get only two orders and third order is not possible.

Q5. What is main difference between a prism spectrum and a grating spectrum?

Ans. In grating spectrum red colour is deviated most and violet least while this order is reversed in prism spectrum.

Q6. Why is the prism spectrum more intense than the grating spectrum?

Ans. In case of prism, the light is concentrated in one spectrum while in case of grating, the incident light is diffracted into spectra of various orders moreover most of the light is concentrated in direct image where no spectrum is formed.

Q7. What is the effect of increasing the number of lines per cm of the grating on the diffraction pattern?

Ans. Increasing the number of lines per cm decrease the grating element $(a+b)$. As a result the angle of diffraction θ increases for a given order. This results in a less number of spectra separated by large angles or an increased dispersive power.

Q8. What type of grating is used in this experiment?

Ans. Plane transmission grating.

Q9. What will happen if the slit is illuminated with white light?

Ans. The central band will be white in the other spectra red will be in the outermost position and violet will be in the innermost position.

Q10. What happens if the ruled surface of the grating faces the collimator?

Ans. The rays will be diffracted and then undergo refraction while passing through the glass plate. Therefore, the measured θ will not be due to diffraction alone.

Q11. What are the uses of a diffraction grating?

Ans. It helps to study diffraction spectra and measure wavelength.

Q12. In the experiment on diffraction grating, what kind of diffraction occurs and how?

Ans. Fraunhofer's diffraction occurs since the spectrometer is focused for parallel rays, so that the source and the images are effectively at infinite distance from the grating.

EXPERIMENT No. 4

Object: To determine the refractive index of the material of the prism using spectrometer.

Apparatus: A spectrometer, an electric lamp, glass prism, a magnifying, an electric lamp and a source of light(Mercury Lamp).

Theory and Formula: Let a narrow parallel beam of light coming from the collimator fall on the edge A of the prism Fig. 4.1(a). It gets partly reflected from the face AC. If we place a telescope with its axis parallel to KG or AE and image of the slit is seen in it: If we rotate the telescope so that its axis lies parallel to HF or AD, the image of the slit will be seen again in it. The angle of prism A is equal to half the angle Θ through which the telescope is rotated, i.e.

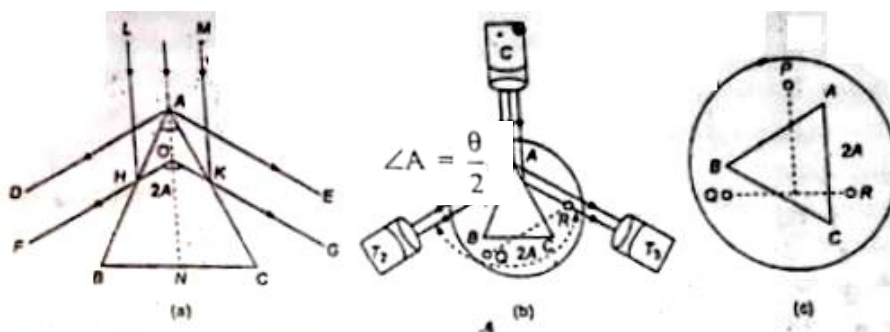


Fig. 4.1 (a) A parallel beam of light falling on the edge of the prism (b) Measuring angle of prism with spectrometer (c) Adjustment of prism on the turn table.

If D_m is the

angle of minimum deviation of a ray refracted through a prism, then refractive index of the material of the prism is given by

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin A/2}$$

Procedure: - Determination of angle of prism

1. Switch on the sodium lamp and let it come to full brightness.
2. Find the vernier constant of the instrument.
3. Adjustment of the telescope and collimator for parallel light.

Set the telescope towards a bright object, say a lamp globe or the sky and adjust the eyepiece so that the cross-wires are clearly visible. Now point the telescope through an open window towards some object say an electric pole about 100 feet away. Adjust the position of eyepiece so that a distinct image of the object is seen and there is no parallax between this image and that of cross wires. The telescope now turned towards the collimator with its slit illuminated with a monochromatic source of light say a sodium lamp. The collimator slit is then adjusted until a distinct image of the slit falls on the cross-wires. The instrument is now adjusted so that parallel rays pass from the collimator to the telescope.

4. Place the prism on the table of the spectrometer so that its refracting edge is vertical and parallel to the slit of the collimator and its face AC should be perpendicular to the line joining the screws Q and R Fig. 4.1(b). Adjust the collimator to illuminate both the faces bounding the edge A.
5. Adjustment of the prism on turn table

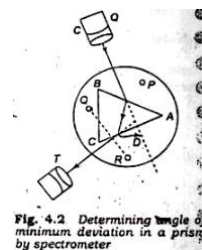
Make the table horizontal by placing a spirit level on it in two perpendicular directions, parallel to QR and perpendicular to QR. Fig. 4.1(c). Place the prism with its reflecting face say AC normal to one of the sides say QR of the triangle formed by the three screws. Rotate the turn-table so that the light from the collimator falls on this face of the prism and is received in the telescope after reflection. If the face is vertical, the image of the slit will appear to be in the same part of the field of view of the telescope as seen directly. If not, then the three screws should be adjusted to restore the image in the centre. Now rotate the table to receive the reflected light from the other face AB bounding the refracting edge. If the image is not in the centre now, then adjust with the help of screw P only as it will not disturb the previous adjustment because it will not turn the face perpendicular to QR out of its vertical plane. This process is repeated till the image reflected from both the reflecting faces is symmetrical with respect to the cross wires.

6. Rotate the telescope (keeping the prism table fixed) to the position T_1 so that the rays from face AB enter the telescope so that the image of the slit lies in the centre of crosswire. To make fine adjustments, the telescope is clamped and tangent screw attached to the telescope is used. Note the readings on both the vernier scales V_1 and V_2 .
7. Now rotate the telescope to the other position F, so that the light rays reflected from face AC enter the telescope and the image of the slit lies in the centre of cross-wires. Note the readings on both the vernier scales V_1 and V_2 . The difference between two positions of the vernier gives twice the angle A of the prism. Take at least 3 observations and record them in a tabular form.

8. Angle of minimum deviation

Place the prism on the prism table in such a way that its centre coincides with the centre of the table and its reflecting face say AC is perpendicular to the line QR. Rotate the table so that the light from the collimator falls on the face AB at grazing incidence so that after refraction it emerges out of the face AC. (Fig. 4.2). Rotate the prism table slowly and see the image of the slit through this face. When the image becomes stationary and tends to move in the opposite direction, fix the prism table. Rotate the telescope so that the image of the slit is seen through it and make final adjustment by rotating the prism table gradually so that the image remains stationary when it is slightly rotated. Fix the prism table again and adjust the position of the telescope so that the image lies in the centre of cross wires. This is the minimum deviation position. Note the readings on both the vernier scales V_1 and V_2 .

9. Keeping the table fixed, remove the prism. Release the telescope and turn to see the slit directly. Adjust the position of telescope so that the slit lies the centre of cross wires symmetrically. Repeat to take three sets of observations as shown in the table.



it
in

Observations

Value of 1 circular scale division =

Vernier constant of the spectrometer =

S. No.	Vernier Scale Reading MSR + (VSR x LC)		Difference θ (degree)	Mean θ_{av} (degree)	Prism Angle A (degree)
	Telescope on LHS (light through left Face of Prism)	Telescope on RHS (light through right Face of Prism)			
1	V1 = V2 =	V1 = V2 =	$\theta_1 =$ $\theta_2 =$	$\theta_{av} =$	A =

Line	Vernier Scale Reading MSR+(VSRxLC)		Difference θ_{dev} (degree)	Angle of Min Dev θ_m (degree)
	For minimum deviation ray	For Direct ray		
Violet	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_v =$
Indigo	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_i =$
Blue	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_B =$
Green	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_G =$
Yellow	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_Y =$
Orange	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_O =$
Red	V ₁ = V ₂ =	V ₁ = V ₂ =	$\theta_1 =$ $\theta_2 =$	$\theta_R =$

Calculations:

Angle of deviation for violet =.....

$$\mu_{\text{violet}} = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \dots\dots\dots$$

Result: The refractive index of material of prism.

Percentage Error :- Refractive index (Standard Value) of the material of given prism from table=

$$\text{Percentage Error} = \frac{\text{Difference of standard and experimental result}}{\text{standard result}} \times 100$$

Precautions:

1. In finding the angle of the prism, the prism-table should be kept throughout.
2. If vernier pass the zero of the scale as telescope is rotated, this face should be taken into account in determining the angle turned.
3. Eyepiece should be adjusted so that the cross-wires are clearly visible.
4. Ensure that the plane of the prism table is horizontal and the axes of the telescope and collimator lie in the same horizontal plane.
5. The slit of collimator should be narrow and adjust its position with respect to the collimator lens so that the image of the slit is sharply focused.
6. Both the vernier scales should be read to avoid error due to eccentricity.
7. The telescope should be focused for infinity and collimator is adjusted to give a parallel beam of light.

Sources of Error

1. Backlash error may arise during rotation of telescope or prism table.
2. Axes of telescope and collimator may not lie in the same horizontal plane.
3. The telescope may not be properly set for infinity.
4. The prism table may not be exactly horizontal.

VIVA-VOCE QUESTIONS

Q1. Why should the prism be placed with the refracting edge A at the center of the prism table ?

Ans. By doing so, (i) the maximum amount of light available from the collimator enters the telescope and brightness of the image increases, (ii) the resolving power of the instrument increases because the full length of the base of prism is utilised.

Q.2. Define refractive index ?

Ans. Refractive index of the denser medium w.r.t. rarer medium is defined as the ratio of sine of angle of incidence in the rarer medium to the sine of angle of refraction in the denser medium i.e. $(\sin i / \sin r)$

Q.3. Define refractive index in terms of velocity of light ?

Ans. It may be defined as the ratio of velocity of light in vacuum to the velocity of light in the medium.

Q.4. Does the refractive index change with colour or wavelength of light ?

Ans. Yes. Refractive index varies inversely as the square of wavelength of light. It is greater for violet rays (which have shorter wavelength) and small for red rays (which have longer wavelength).

Q.5. What is a sodium lamp ?

Ans. A sodium lamp is a source of monochromatic light.

Q.6. What is a monochromatic light ?

Ans. Light of single colour (or wavelength) is called monochromatic light.

Q.7. What is the colour of light obtained from sodium lamp ?

Ans. Two yellow lines D₁ and D₂, of nearly the same wavelength. i.e. 5890 Å and 5896 Å are obtained.

Q.8. Is sodium lamp a monochromatic source ?

Ans. In strict sense no, but practically yes. It contains light of two wavelengths 5896 Å and 5890 Å. As these two are very close to each other that is why it is called monochromatic source.

Q.9. Why red colored light is seen first when we switch on sodium lamp and then turns into yellow ?

Ans. On switching on the sodium lamp, the discharge first passes through the neon gas which gives its characteristic red colour. The heat produced causes the metallic sodium specks to evaporate and sodium vapors are formed. Discharge then passes through sodium vapors and golden yellow light is emitted.

Q.10. Define angle of deviation ?

Ans. The angle between direction of the incident ray falling on one refracting face and the direction of the emergent ray coming out of the second refracting face of the prism is called angle of deviation.

Q.11. What is angle of minimum deviation ?

Ans. The angle of deviation for which the ray emerging out of second refracting face of the prism becomes minimum and the image just becomes stationary on rotating the prism table, is called angle of minimum deviation.

Q.12. How does deviation angle depend upon the angle of prism?

Ans. Deviation suffered by a ray of light on passing through prism is given by $\delta = (\mu - 1)A$. For a given colour and material of the prism, the angle of deviation δ is directly proportional to the angle of prism.

Q.13. Does deviation depends upon refractive index of material?

Ans. Yes, for a particular angle of the prism and a particular wavelength, the angle of deviation increases with the increase of the refractive index of the material of the prism.

Q.14. When does an incident ray of light suffer minimum deviation?

Ans. The angle of deviation is minimum when the ray passes through the prism symmetrically i.e. when the incidence is equal to the angle of emergence.

Q.15. Which colour in the spectrum deviates more and why?

Ans. Deviation angle is inversely proportional to the square of wavelength. The deviation of red ray is least where that of violet is maximum.

Q.16. Why should the prism be placed in minimum deviation?

Ans. Because the image is brightest in this position and the spectrum obtained will be pure.

Q.17. What is a prism?

Ans. A prism is made up of a transparent material, bounded by two plane faces, inclined to each other at an angle. The angle between these two faces is called the angle of prism.

Q.18. What is a spectrometer?

Ans. An instrument used to observe and study the spectrum of a given source is called spectrometer

Q.19. Name the different parts of a spectrometer

Ans. (1) Collimator (i) a prism table (ii) a telescope

Q.20. Why do you take readings from both the vernier?

Ans. To eliminate the error arising out of the fact that the axis of rotation of the prism table or the telescope may not coincide with the centre of the circular main scale.

Q.21. How can you check whether the readings noted from the two verniers are correct?

Ans. The difference between the readings of the two verniers should be nearly 180° .

Experiment 5

Objective: To determine the dispersive power of the prism using spectrometer and mercury source.

Apparatus: Spectrometer, prism, a spirit level, a magnifying lens, mercury lamp and sodium lamp.

Theory: The angular dispersion is defined as the difference between the deviations produced in the extreme colours of the spectrum.

It D_v and D_r are the deviations for violet and red colours respectively, then

$$\text{angular dispersion } \alpha = D_v - D_r = (\mu_v - 1)A - (\mu_r - 1)A$$

$$= (\mu_v - \mu_r)A$$

where μ_v and μ_r are refractive indices of violet and red light, A is the angle of prism.

The *mean deviation* is the deviation produced for the mean colour for the spectrum. The mean colour is generally taken as the yellow colour :-

$$\text{mean deviation} = (\mu_y - 1)A$$

where μ_y is the refractive index of yellow light.

The dispersive power of the material of a prism is defined as the ratio of angular dispersion to mean deviation.

$$\text{dispersive power } \omega = \frac{(\mu_v - \mu_r)A}{(\mu_y - 1)A} = \frac{\mu_v - \mu_r}{\mu_y - 1}$$

The dispersive power depends upon the nature of the material of the prism and is independent of the angle of the prism.

Procedure :

1. Make all necessary adjustments of the spectrometer and level the prism table as discussed in experiment 4.
2. Place the collimator of the spectrometer in front of the sodium lamp and determine the angle A of the prism as done earlier (Expt. 4)
3. Using mercury lamp find the angles of minimum deviation D_m for violet, yellow and red lines following the procedure discussed in previous experiment.
4. Calculate the value of μ_v , μ_y and μ_r using the formula

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Observations

Vernier constant of the spectrometer =

Angle of Prism, A

S. No.	Reading of vernier V ₁			Reading of vernier V ₂		
	For Position of Telescope Through		Difference (θ)	For Position of Telescope Through		Difference (θ)
	First face	Second face		First face	Second face	
1						
2						
3						

Mean value of θ =

Angle of the prism, A = θ/2 =

Angle of minimum deviation (D_m)

S. No.	Line	Vernier Reading	Minimum Deviation Ray	Direct Ray	Difference D _m
	Violet				Mean D _v
	Yellow				Mean D _y
	Red				Mean D _r

Calculations : Calculate the refractive index of violet, yellow and red colours from the expression

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Putting values of D_v, D_y, and D_r, we get value of μ for three colours as

$$\mu_v = \dots\dots\dots$$

$$\mu_y = \dots\dots\dots$$

$$\mu_r = \dots\dots\dots$$

$$\text{Dispersive power, } \omega = \frac{(\mu_v - \mu_r)}{(\mu_y - 1)} = \dots\dots\dots$$

Dispersive power of the material of prism =(given)

Precautions and Source of Error Same as in experiment 4.

VIVA-VOCE QUESTIONS

Q1. For which colour μ of the material of prism is () minimum (i) maximum

Ans. is minimum for red colour and maximum for violet colour.

Q.2. Do the light rays of different colour travel with the same velocity while passing through the prism?

Ans. No. In vacuum, light waves of all colours move with the same velocity. But in prism violet ray travels with a slower velocity than the red and hence the refractive index for the violet colour is greater than that for the red.

Q.3. What do you mean by dispersion? Is it a constant for a prism?

Ans. The angular separation produced between two colours by a prism is called dispersion. It depends on the angle of incidence, angle of prism, the μ for the prism and on the wavelength of the incident light. It is not a constant quantity.

Q.4. Which device, other than a prism, disperses light?

Ans. Diffraction grating also disperses light.

Q.5. Define dispersive power?

Ans. The dispersive power of a material is its ability to disperse various colours constituting the incident light. For the prism, the dispersive power is defined as the ratio of the angular dispersion to the mean deviation. The dispersive power of the grating is defined as the change in the angle of diffraction corresponding to a unit change in the wavelength of light. (Refer to Sec. 2.7 of Chapter 2)

Q.6. On what factors does the dispersive power of a prism depend?

Ans. Dispersive power depends on the material of the prism.

Q.7. Can you find the dispersive power of a prism using sodium light?

Ans. No, because it has only two yellow lines.

Q.8. What is the unit of dispersive power? What is its dimension?

Ans. No units, as it is the ratio of two similar quantities. It has no dimension.

Q.9. Which prism is preferable, one made of flint glass or that made of crown glass? Why?

Ans. The prism made of flint glass is preferred as it gives a greater dispersion compared to a crown glass prism.

Q.10. Why is the spectrometer telescope fitted with Ramsden's eye-piece and not Huygen's eye piece?

Ans. The telescope with a Huygen's eye-piece does not permit any measurement since the cross-wire cannot be conveniently used with this eye-piece, whereas with Ramsden's eye piece a cross-wire can be used for measurements.

Experiment 6

Objective: To determine the specific rotation of cane sugar solution with the help of half shade polarimeter.

Apparatus: Laurent's polarimeter and half shade device, sodium lamp, 50 cc flask, beakers and analytical balance.

Experimental Set-up: Students should carefully read Art. 3.14 of chapter 3 and reproduce here the experimental set up shown in Fig. 3.22.

Theory: *Specific rotation* is the rotation produced by one decimeter (10 cm) length of the solution of unit concentration.

Thus, if x gms of sugar are dissolved in V cc of the solution and the length of the tube containing the solution is l decimeter and θ is the rotation produced, then the specific rotation, S of cane sugar at the given temperature and corresponding to the wavelength λ is given by

$$S = \frac{\theta/l}{x/V} = \frac{\theta}{lC} \quad \text{where } C = \frac{x}{V} \text{ is the concn, in gm/c.c.}$$

If l is in centimeter, then

$$S = \frac{10\theta}{lC}$$

Procedure:-

1. Prepare 20% solution of sugar. For this, take a clean dry beaker and weigh it. Add 20 gm of sugar in it and weigh again. First dissolve the sugar in 50 c.c. of water. Add more water gradually by a pipette till the sugar dissolves completely. Shift the solution into measuring cylinder. Rinse the beaker with a small quantity of water and add to the cylinder. Add more water with a pipette to make the total volume of 100 c.c.
2. Filter the solution in a clean beaker.
3. Clean the polarimeter tube, fill it with water and slip the window gently on the tube taking care that no air bubbles are left inside the tube. Screw the cap gently.
4. Put the tube in position in the Laurent's half-shade polarimeter. Rotate the analyser N_2 by rotating the circular scale till the field of view is completely and uniformly dark. In this position, both the halves of the half shade device are in equally dark position.

- Read the position of the analyser on the circular scale. Get a second position of extinction by rotating further through 180° . Note the reading again.
- Now fill the tube with 20% sugar solution. Place it in position. The two halves of the field appear unequally dark. Rotate the analyser and note the scale readings for two equally dark positions. The difference of the corresponding scale readings with solution and water gives the rotation produced (θ). Take the mean of the two θ 's.
- Add 50 c.c. of water to 50 c.c. of solution and mix thoroughly. This gives 10% solution.
- Repeat the step 6 with 10%, 5%, 2.5% and 1.25% solution and take the readings.
- Record the temperature of the solution and report it with the result.

Observation :

Weight of the beaker :.....gm
 Weight of the beaker + sugar =.....gm
 Weight of sugar =.....gm
 Length of the tube l =.....cm.
 Temperature of the solution =..... $^\circ\text{C}$

Table for θ :

Vernier constant of the polarimeter =

Scale reading through water

Ist position x_0 =
 IInd position x_0 =

S. No.	% Solution (gm per c.c.)	Conc. C (gm/cc)	Scale reading through solution		Rotation		Mean θ (deg)
			1 st position x_1	2 nd position x_2	1 st position $\theta_1 = x_1 - x_0$	2 nd position $\theta_2 = x_2 - x_0$	
1	20	0.2					
2	10	0.1					
3	5	0.05					
4	2.5	0.025					
5	1.25	0.0125					

Calculations : Plot a graph between concentration C and rotation θ (Fig. 6.1). It will be a straight line.

Slope of the graph is $\frac{\theta}{C} = \dots\dots\dots$

Specific rotation = $\frac{10\theta}{lC} = \frac{10}{l} \times \text{slope}$

Result : The Specific rotation of sugar solution is $^\circ\text{C}$ with $\lambda = \dots\dots^\circ\text{A}$.

Precautions and Sources of Error

1. There should be no air-bubbles in the polarimeter tube.
2. The solution should be filtered in order to make it dust free. Moreover, a pure variety of sugar should be employed otherwise there will be unnecessary diminution of intensity of light.
3. While screwing the cap of the tube, do not apply too much pressure, otherwise the glass end is likely to be strained and this will produce spurious rotation.
4. The analyzer should be set in correct position with respect to polarizer. It should be set again in a position 180° apart. During one set of observation, the position of the polarizer should not be disturbed.
5. Since specific rotation of an active material varies with temperature, the latter should be recorded and reported with the result.
6. The readings should be taken in equally bright position and not in equally bright position.

VIVA-VOCE QUESTIONS

Note: Students are referred to short answer questions on polarization on page 118 to 120.

Q.1. What are uniaxial and biaxial crystals?

Ans. The crystals having one direction (optic axis) along which the two retracted rays travel with the same velocity are called as uniaxial crystals. In biaxial crystals, there are two optic axes.

Q.2. What do you mean by optical activity, optical rotation-and angle of rotation ?

Ans. The property of rotating the plane of vibration of plane polarized light about its direction of travel by some crystal is known as optical activity. This phenomenon is known as optical rotation and the angle through which the plane of polarization is rotated is known as angle of rotation.

Q.3. What is specific rotation ?

Ans. The specific rotation of a substance at a particular temperature and for given wavelength of light may be defined as the rotation produced by one decimeter length of its solution when concentration is 1 gm. per c.c. Thus

$$\text{Specific rotation} = \frac{\theta}{l \times c}$$

where θ is angle of rotation in degrees, l the length of solution in decimeter and C , the concentration of solution in gm. per c.c.

Q.4. What is a polarimeter ?

Ans. It is an instrument used for measuring the angle of rotation of the plane of polarization by an optically active substance.

Q.5. What is a saccharimeter?

Ans. A polarimeter used in case of sugar analysis is called saccharimeter.

Q.6. What do you mean by dextro and laevo-rotatory substances?

Ans. When the optically active substance rotates the plane of polarization of light towards right, is called as right handed or dextro-rotatory. If the substance rotates the plane of polarization towards left, it is called as left handed or laevo-rotatory.

Q.7. What is half shade or a Laurent's plate?

Ans. The Laurent's half shade plate consists of a semicircular half wave plate of quartz cut parallel to optic axis so that it introduces a phase change of π between extra ordinary and ordinary ray passing through it and semi-circular glass plate with thickness such that it absorbs the same amount of light as quartz plate. Both are cemented along diameter. It is prepared to suit for one particular wavelength.

Q.8. Where is the half-shade plate fitted in the polarimeter?

Ans. This is fitted between the polarising Nicol and the polarimeter tube containing the solution.

Q.9. What is an optically active substance?

Ans. If the plane of polarization of a polarised light is rotated in passing through a substance, it called an optically active substance.

Q.10. Why is the plane of polarisation rotated in optically active substances?

Ans. when plane polarised light passes through an optically active substance, it breaks into two beams which are circularly polarized in opposite directions and have unequal velocities. When these beams emerge out of the substance, they recombine. This again gives plane polarized light but the resultant direction of vibration becomes different from the initial one. This results in the rotation of the plane of polarization.

Q.11. What are the factors on which the specific rotation of a solution depends?

Ans. (i) On wavelength of light used, (ii) on temperature of the solution, and (iii) on the nature of the solvent and the solute.

Q.12. How does the specific rotation of cane-sugar vary with temperature ?

Ans. Specific rotation of cane-sugar S at any temperature T is given by the relation

$$S = 66.5^\circ - 0.0184 (T - 20^\circ)$$

Thus, when $T = 20^\circ\text{C}$,

$$S = 66.5^\circ$$

Q.13. On what factors does the angle of rotation of the plane of polarization depend?

Ans. (i) On the length of the liquid column through which light passes ($\theta \propto l$). (ii) on concentration of the solution ($\theta \propto c$), (iii) on wavelength of light used and (iv) on temperature of the solution.

EXPERIMENT NO. 7

Object: To find the wavelength of He-Ne LASER source using transmission grating.

Apparatus: He-Ne Laser, Diffraction grating, optical bench with four stands, a screen with a graph paper pasted on it, metre-scale and a travelling microscope.

Theory : When a monochromatic beam of wavelength λ is diffracted by a diffraction grating, the n th order principal maxima is formed at an angle θ_n given by

$$(a+b) \sin \theta_n = n\lambda$$

where $(a+b)$ is the grating constant and

$$\lambda = \frac{(a+b) \sin \theta_n}{n}$$

When laser light is incident on the diffraction grating and diffraction pattern is obtained on the screen formed by a graph paper, $\sin \theta_n$ can be obtained for different order maxima as shown in Fig. 7.1.

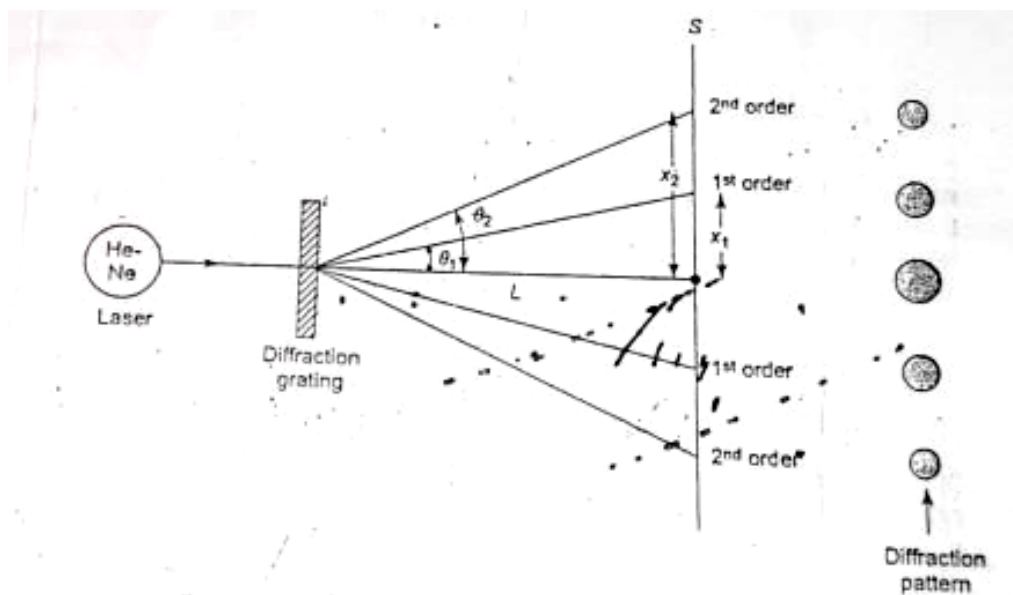


Fig. 7.1

From the figure, we have

$$\sin \theta_n = \frac{x_n}{\sqrt{L^2 + x_n^2}} = \frac{x_n}{L} \quad \text{if } L \gg x_n$$

Substituting $\sin \theta_n$ for different orders in the above equation, λ can be obtained.
The experimental set-up is shown in Fig. 7.2

Procedure :

1. Mount a diffraction grating on a stand and illuminate it with the laser beam coming from the He-Ne Laser source. Place a graph paper pasted on a screen S vertically at a distance of 1 to 2m from the grating. Adjust the distance of screen from the grating till a sharp diffraction pattern in the form of several bright spots of diminishing intensity is obtained. Mark the centre of the spots(maxima) with a pencil.
2. Measure the distance L between the grating and the screen with a meter scale. Also measure the distance of maxima x_n from the central maximum with the help of the scale on the graph-paper or a travelling microscope.
3. Remove the diffraction grating and measure the diameter d of the direct spot on the screen kept at a distance D from the laser source.
4. Repeat step 3 for different distances of the screen from the laser.
5. Calculate the angular spread θ

Observations

(i) Measurement of λ

Number of lines per inch on the grating, $N = \dots\dots\dots$

$$\text{Grating constant} \quad (a + b) = \frac{1}{N} = \dots\dots \text{cm}$$

Distance of screen from the grating, $L = \dots\dots\dots \text{cm}$

S. No.	Order of diffraction	Distance of maxima from central point, x_n (cms)	$\sin \theta_n = \frac{x_n}{\sqrt{L^2 + x_n^2}}$	$\lambda = \frac{(a + b) \sin \theta_n}{n}$

Mean $\lambda = \dots\dots\dots \text{cm}$

$= \dots\dots\dots \lambda$

Result: The wavelength of He-Ne Laser light $\lambda = \dots\dots\dots A^0$
Actual value = 6328 A
% error =

Precautions and Sources of Error

1. Do not see at the laser source directly as it may cause damage to the eye.
2. The graph used at the screen should be vertically straight
3. The diffraction grating should be very near to the laser source

VIVA-VOCE QUESTIONS

Note : Should thoroughly read the short answer questions at pages 147 to 150 of chapter 4.

Q.1. In He-Ne laser lasing is through neon gas. What is then the role of helium?

Ans. In He-Ne laser, helium serves as a buffer gas and helps in the excitation of neon atoms. Since He-atoms are much lighter than Ne-atoms, electrons transfer energy more readily to He-atoms. The excited He-atoms in turn transfer their energy to Ne-atoms.

Q.2. Why is it necessary to use a narrow tube in a He-Ne laser ?

Ans. A narrow discharge tube is necessary for the rapid de-excitation of atoms by collision with the walls. With a tube of large diameter, the probability of collisions of atoms with the walls decreases and less atoms are available at the ground level for further excitation. This can cease the laser action in due course.

EXPERIMENT NO. 8

Objective: To determine the numerical aperture (NA) of an optical fibre.

Apparatus: Optical-fibre, two fibre optic chucks, He-Ne laser, a microscopic objective and a screen with a graph paper pasted on it.

Theory: Students are advised to go through chapter 5 on fibre optics. Numerical aperture (NA) of the optical fibre is a measure of light gathering ability of the fibre and is an important parameter which determines the angle of the "light acceptance cone") at the input end of the fibre. The numerical aperture is defined as the sine of half angle of the acceptance cone. It is given as

$$NA = \sin \Theta_a = \sqrt{\mu_f^2 - \mu_c^2}$$

Where μ_f and μ_c are refractive indices of cord and cladding respectively. Thus to determine the numerical aperture, the acceptance angle Θ_a of the fibre should be known.

In a short length of optical fiber, ideally a ray entering at an angle i at the input end comes out at the same angle i from the output end. Therefore, the emerging rays from the output end of the fiber will also appear as a cone of semi-angle i . It is thus easy to make measurements on this end of the fiber to determine the numerical aperture of the fiber. If the emerging rays from the output end of the optical fiber make a spot of diameter D on a screen kept at a distapce L from the output end of the fiber, then as obvious from Fig. 8.1(b),

$$\sin \theta_a = \frac{D/2}{\sqrt{(D/2)^2 + L^2}} = \frac{D}{\sqrt{D^2 + 4L^2}}$$

$$NA = \frac{D}{\sqrt{D^2 + 4L^2}}$$

Procedure:

1. Insert both the ends of the optical fibre in the fibre optic chucks mounted in the uprights on an optical bench as shown in the figure 8.1(a).

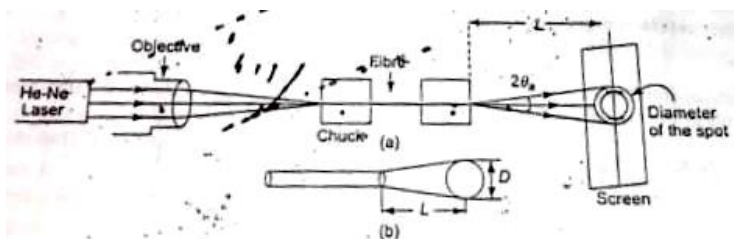


Fig. 8.1 Setup of optical fibre for measuring NA

2. Switch on He-Ne laser.

3. Adjust the He-Ne laser so that laser beam is coupled into one of the fibre ends through a 20 x microscopic objective mounted in an upright.
4. Fix a graph sheet on the screen mounted in an upright.
5. Place the screen at some distance at the other end of optical fibre (i.e. output end) such that it is perpendicular to the axis of the fibre.
6. Adjust the position of the screen by moving it towards or away from the output end of fibre so that a bright circular spot is formed on the screen.
7. Measure the diameter of the circular spot formed on the screen. Let it be D. Also measure the distance between the output end of optical fibre and screen. Let it be L.
8. Repeat the above experiment to take at least four observations for different values of L and D.

Observations

S. No.	Distance between output end of optical fibre and screen L cm	Diameter of circular spot D cm	$NA = \frac{D}{\sqrt{D^2 + 4L^2}}$

Mean value of numerical aperture =

Result: Numerical aperture =.....

Precautions:

1. Optical fibre should not bend when the light is being propagated.
2. Light should be coupled properly into the fibre so that a bright spot of light is obtained.
3. Losses due to bending, leakage, dispersion etc. should be minimised.
4. We should ensure accurate measurement of diameter of the spot.
5. Optical fibre axis should be parallel to optical bench.

Sources of Error

1. There may be losses due to bending, radiation and leakage etc.
2. The edges or ends of the optical fibre might not be sharply cut and the bright spot may not be exactly circular.
3. The intensity of light is reduced as the distance between the screen and output end of fibre is increased.

VIVA-VOCE QUESTIONS

Q.1. What is an optical fibre?

Ans. An optical fibre is a cylindrical structure which consists of a glass core surrounded by a cladding of slightly lower refractive index. It is the light equivalent of microwave wave guide having a very wide band width.

Q.2. Give an approximate general idea of core and cladding diameter?

Ans. The core diameter varies between 5 μ m to about 100 μ m whereas cladding diameter is usually of the order of 125 μ m.

Q.3. How is fibre protected from damaging?

Ans. To give, greater strength and protection to the fibre, usually, a soft plastic coating of about-250 μ m diameter surrounded by another layer of coating is made.

Q.4. Which material is generally used in present day communication grade fibres?

Ans. Silica (SiO₂) based fibres are generally used.

Q.5. What is a step index fibre?

If the core of optical fibre has uniform refractive index, it is called a step index fibre.

Q.6. What do you mean by graded index fibre?

Ans. If the core of optical fibre has a non-uniform refractive index, gradually decreasing from the centre towards the outer region, it is called a graded index fibre.

Q.7. How is the propagation of light taking place in an optical fibre?

Ans. When light enters at one end of the fibre under required conditions, most of light will propagate or move down the length of fibre and exit from the far end whereas a small part of light will be absorbed or scattered by the cladding. Light enters one end of the fibre at a slight angle to the axis. As it encounters a surface with a different refractive index, it suffers total internal reflection and gets propagated to emerge out of the fibre.

Q.8. What are the conditions for total internal reflection?

Ans. The following conditions should be fulfilled for total internal reflection to take place at the fibre walls:

1. The core must have a slightly higher refractive index than the cladding..
2. Light must have an angle of incidence greater than the critical angle.

The light so reflected bounces off the walls after repeated total internal reflections and proceeds in a zig-zag path.

Q.9. What are the advantages of optical fibres in a communication systems?

Ans. Optical fibres offer some important advantages over wires :

1. Light travels faster in glass than electrical signals in the conventional metallic wires.
2. By using monochromatic light (say lasers), the signal distortion is low.
3. Optical fibres have a high band width (and hence data carrying capacity) of the order of GHz.
4. Optical fibres can carry 100 million times more information and 100 times faster than telephone lines.
5. Optical fibres are very light in weight, easily twistable and have a low attenuation i.e. 0.5 decibels which is approximately 10 times less than telephone cables.

Q.10. Is crosstalk possible in optical fibre?

Ans. There is no possibility of cross-talk because an optical fibre is made up of cylindrical silica which is non-conductive and non-radiative.

Q.11. Does the high temperature damage the fibres?

Ans. Fibres are resistive to high temperature as the melting point of silica is very high i.e. 1900 degree Celsius.

Experiment 9

Objective: To plot a graph between distance of the knife edge from the centre of gravity and time period of bar pendulum. From the graph, find (a) the acceleration due to gravity (b) the radius of gyration.

Apparatus: Bar pendulum, a small metal wedge, a spirit level, a telescope, a stop watch, a metre-rod, a rigid support fixed on the wall.

Theory: A bar pendulum consists of a uniform metal bar having holes drilled along its length symmetrically on either side of the centre of gravity at intervals of about 5 cm. Two knife edges are placed symmetrically with respect to C.G. as at A and B; Fig. 9.1. The bar is allowed to oscillate about a horizontal knife-edge passing through each of holes in turn, beginning with that nearest to one end and the time period is noted in each case. If a graph is plotted between the distance of the knife-edges from the centre of gravity and the corresponding time period T , it has the shape shown in fig.9.2

If a horizontal line ABCDE is drawn, it cuts the graph in points A,B,D and E about which the time period is the same. For A as the centre of suspension, D is the centre of oscillation and for E as the centre of suspension, B is the centre of oscillation. The distance AD and BE thus give the length of the equivalent simple pendulum L . If T is the corresponding time period T , then

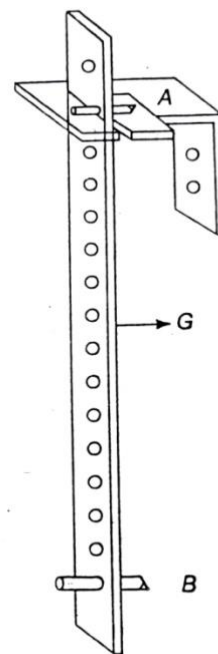


Fig. 9.1 Bar pendulum

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$g = \frac{4\pi^2 L}{T^2}$$

The radius of gyration

$$k = \sqrt{ll'}$$

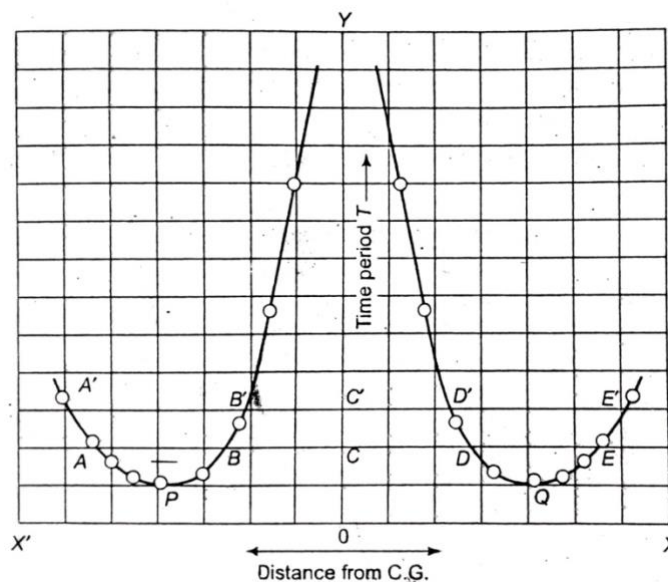
where

$$l = \frac{AC+CE}{2} \text{ and } l' = \frac{BC+CD}{2}$$

Procedure:

1. Balance the bar on the sharp wedge and mark its C.G. at G.
2. Fix the knife edges in the outermost holes on either end of the bar parallel to each other with their sharp edges pointing towards G.

3. Check with a spirit level that the glass plates G fixed on the rigid bracket on the wall are horizontal. Suspend the pendulum by placing the knife edge on the side A perpendicular to the slot in the bracket and see that the pendulum hangs vertically.
4. Adjust the eye-piece of the telescope so that the cross wires are clearly visible through it. Focus the telescope on the bar from about one metre distance and see that the point of intersection of the crosswire coincides with the line AB or a reference line drawn on the pendulum.
5. Displace the bar slightly and release it. It will begin to vibrate. Note the time for 20 vibrations and also measure the length from the point G upto the bottom of the first hole, i.e., upto the point about which the pendulum turns. Distances can also be measured from one end of the bar.
6. Suspend the pendulum on the knife edge on B side of the pendulum and take the observation as in step-5 above.



7. Now insert the knife edges in holes, 2, 3.. on either side of the C.G. and note the time for 20 vibrations in each case and measure the corresponding distances from G.
8. Plot a curve between distance from C.G. and the corresponding time period T (time for one vibration).
9. Draw two horizontal lines ABCDE and A'BCD'E and do the calculations.
10. Find the mass of the pendulum to determine the moment of inertia.

Observations

Least count of stop watch = ... s

Vernier constant of callipers = .. cm.

	No	Distance from G (cm)	Time for 20 vibs	Time period T(s)
(1) For the holes on side A	1			
	2			
	3			
				

(ii) For the holes on side B	1			
	2			
	3			
				

Calculations: (1) Plot a graph between from C.G. and T as shown in Fig. 9.2

(A) For Acceleration due to Gravity g:

(i) From the line ABCDE

$$L = \frac{AD+BE}{2} = \dots \text{ cm}, \quad T = \dots \text{ sec}$$

$$g = \frac{4\pi^2 L}{T^2} = \dots \text{ cms}^{-2}$$

(ii) From the line A' B' C' D'

$$L = \frac{A'D'+B'E'}{2} = \dots \text{ cm}, \quad T = \dots \text{ sec}$$

(B) For Radius of gyration k

(i) From the line ABCDE in the graph

$$l = \frac{AC+CE}{2} \text{ and } l' = \frac{BC+CD}{2}$$

$$k = \sqrt{ll'} \dots \text{ cm}$$

(ii) From the line A' B' C' D'

$$l = \frac{A'C'+C'E'}{2} \text{ and } l' = \frac{B'C'+C'D'}{2}$$

$$k = \sqrt{ll'} = \dots \text{ cm}$$

(iii) Also $k = \frac{PQ}{2}$ or the mean length corresponding to the minimum time period.

$$= \dots \text{ cm}$$

$$\text{Mean } k = \dots \text{ cms} = \dots \text{ m}$$

Result:

1. The acceleration due to gravity $g = \dots \text{ ms}^{-2}$
2. Radius of gyration = $\dots \text{ m}$

(2) Second Method for calculation of g and k

$$T = 2\pi \sqrt{\frac{\left(\frac{k^2}{l} + l\right)}{g}}$$

$$T^2 = \frac{4\pi^2}{g} \left(\frac{k^2}{l} + l\right)$$

$$T^2 l = \left(\frac{4\pi^2}{g}\right) k^2 + \left(\frac{4\pi^2}{g}\right) l^2$$

So if a graph is plotted between l^2 and T^2 (Fig. 9.3), the slope of the graph gives $\frac{4\pi^2}{g}$ and negative intercept on l axis gives k^2 .

Hence both g and k can be determined by this method simultaneously.

Precautions and Sources of Error

1. The knife-edges should be horizontal and the bar pendulum should hang vertically.
2. Amplitude of vibration should be small ($<5^\circ$).
3. The time of oscillation should be noted after 5 or 6 oscillations so that any irregularities of motion subside.
4. Time should be noted with the help of a good stopwatch using a telescope.
5. The knife edges should be sharp.
6. The support should not be jerky.
7. Smooth and good-sized curve should be drawn.

Weak points: The result may be slightly inaccurate on account of

- (i) curvature of the knife edges,
- (ii) finite amplitude of the swing,
- (iii) yielding of the support and,
- (iv) resistance of air

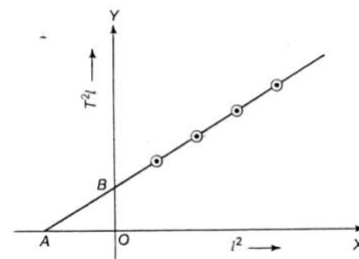


Fig. 9.3

VIVA-VOCE QUESTIONS

Q.1. Define the acceleration due to gravity ' g '

Ans. The acceleration with which a body falls freely towards the centre of earth at a given place under the action of gravity is called acceleration due to gravity and denoted by ' g '.

Q.2. What is the value and unit of ' g '?

Ans. In MKS the value of ' g ' is 9.8 m/sec^2 .

Q.3. Distinguish between acceleration due to gravity and gravitation.

Ans. Every body is attracted towards the centre of earth. This force of attraction of the earth towards its centre is called gravitation and the rate of change of velocity of a freely falling body due to the force of attraction of the earth is acceleration due to gravity,

Q.4. How does the value of ' g ' vary at different places and at different distances from the surface of the earth?

Ans. (i) The value of ' g ' is maximum at the poles and decreases as we move towards the equator where it is minimum (ii) The value of ' g ' is greatest at the surface of the earth and decreases as we go up from the surface or go down inside the surface.

Q.5. What is the effect of altitude, depth and rotation of earth on the value of ' g '?

Ans. Effect of altitude and depth (see Q.4)

Experiment 10

Objective: To determine the Planck's constant.

Apparatus: A vacuum type photoelectric cell mounted inside a wooden box with a wide opening on the side opposite to the cathode, 6 volt DC supply, a voltmeter (0-3 volt range), a rheostat, a sensitive moving coil galvanometer with lamp and scale arrangement, a key, a mercury lamp and few light filters.

Theory: when a photon of energy $h\nu$ is incident on the emissive surface of the cathode, almost all of its energy is transferred to the electron inside the metal. If this energy is greater than the threshold energy W_0 the electron is emitted. W_0 is also called the work function of the metal. Above the threshold frequency, corresponding to the $W_0 (=h\nu_0)$, photoelectrons have a range of energies from 0 to a certain maximum value, and this maximum energy increases linearly with increasing frequency. This is because, out of the total incident energy $h\nu$, a part W_0 is used up as the threshold energy and the rest is stored in the electron as its kinetic energy.

Thus,

$$h\nu = \frac{1}{2}mv_m^2 + W_0$$

$$E_{max} + W_0 \quad \dots(12.1)$$

This eqn. is called the Einstein's Photoelectric Equation.

As W_0 remains constant for a given photoelectric cell, E_{max} varies linearly with the frequency. In this experiment, to find out the maximum kinetic energy E_{max} of the emitted electrons, we reverse bias the photoelectric cell i.e., its anode is made negative. It therefore, repels the emitted electrons and the current decreases. The negative potential on the anode is slowly increased till the stopping potential V_s is reached, when the current stops. When this happens

$$eV_s = E_{max}$$

V_s is called the stopping potential or the cut-off potential because it just stops the electrons from leaving the surface.

From (12.2) and (12.1), we have

$$\begin{aligned} h\nu &= eV_s + W_0 \\ eV_s &= h\nu - W_0 \end{aligned}$$

$$V_s = \frac{h}{e} \nu - \frac{W_0}{e}$$

Thus, if we draw a graph with the frequency ν along the x-axis and the stopping potential V_s along the y-axis, it would be a straight line with slope equal to $\frac{h}{e}$ and negative intercept on y-axis equal to $\frac{W_0}{e}$

Thus the Planck's constant h and the work function W , for a given photoelectric cell can be determined..

Procedure :

1. Make the connections as shown in Fig. 12.1. C and A are respectively the cathode and anode of the photocell.
2. Arrange the mercury lamp just in front of the photocell and set the galvanometer so that the spot moves freely.
3. With no light falling on the photocell, insert the key in K, adjust the voltage in the voltmeter to zero and adjust spot in the galvanometer to zero position. Remove the key.
4. Fit the violet filter in front of the photocell. Switch on the mercury lamp and plug in the key. The galvanometer will show deflection. Adjust the position of the mercury lamp so as to get maximum deflection. Do not move the lamp after this.
5. Increase the potential on anode slowly by the rheostat. The deflection in the galvanometer decreases. Go on increasing the negative potential applied to anode till the spot of light in the galvanometer comes back to zero position. Note: down the potential
6. Repeat the process and take three readings with the same filter.
7. Change the filter one by one from violet to red and repeat the experiment.
8. Make a graph with frequency of the filter ν along x-axis and the stopping potential V , along y-axis.

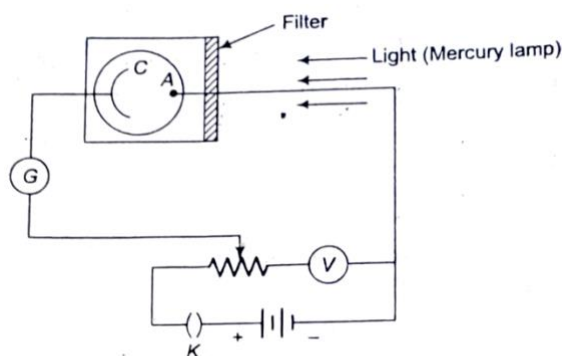


Fig. 12.1 Experimental arrangement for Planck's constant.

Observations

Least count of the voltmeter = 0.05 V

speed of light, $c = 3 \times 10^8$ m/sec

Filter	Violet	Blue	Green	Yellow	Red
Wavelength	4050×10^{-10}	4360×10^{-10}	5460×10^{-10}	5780×10^{-10}	6910×10^{-10}

Frequency $\nu = \frac{c}{\lambda} (\text{sec}^{-1})$					
Stopping Potential 1					
Potential 2					
V_s (volts) 3					
Mean V_s (volts)					

Calculations:

The graph between the frequency ν along x-axis and the stopping potential V_s along y-axis is a straight line as shown in Fig. 12.2. Slope of the line = $\frac{AC}{BC} = \dots$

$$\frac{h}{e} = \text{slope}$$

$$\begin{aligned} \text{Planck's constant } h &= e \times \text{slope} \\ &= 1.6 \times 10^{19} \times \text{slope} \\ &= \dots \text{ J sec.} \end{aligned}$$

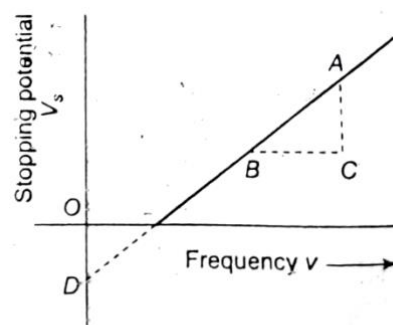


Fig. 12.2

Result: (i) Planck's constant, $h = \dots$ Joule sec
Actual value = $\dots$ Joule sec
% Error = $\dots$ %

(ii) Work function, W_0 for the given photoelectric cell = $\dots$ eV

Precautions and Sources of Error

1. Care should be taken to ensure that when no light is falling on the cathode, the deflection in the galvanometer is zero.
2. The position of the mercury lamp should not be changed during the experiment.
3. Care should be taken to note down the stopping potential. The voltage across the photocell should be increased very slowly and three readings should be taken for each filter.

Weak Points

A sharp cut-off may not be obtained due to ionization of the residual air in the cell and due to secondary emission and photoelectric emission from the anode A.

VIVA-VOCE QUESTIONS

Q.1. Define photoelectric effect.

Ans. It is a phenomenon of emission of electrons from a metal surface when light of suitable wavelength falls on it. The electrons emitted are photo electrons.

Q.2. What is photoelectric cell?

Ans. A photoelectric cell is an arrangement to convert light energy into electrical energy. Photo electric cell are of three types: (i) photoemissive cell (ii) photo-conductive cell (iii) photo-voltaic

Q.3. What is Planck's constant and what is its value?

Ans. The Planck's constant is equal to the ratio of a quantum of energy to the frequency i.e. $h = \frac{E}{\nu}$
Its value is 6.626×10^{-34} Js.

Q.4. Are these methods accurate for the determination of h?

Ans. No, there are several inaccuracies. But give reasonably accurate value of h.

Q.5. What sensitive material is used when the cell is to be used for a visible light?

Ans. The materials are sodium, potassium or cesium on silver oxide or antimony cesium alloy.

Q.6. On what factors, the number of electrons emitted per unit area depend?

Ans. The number of electrons emitted per unit area depend on the intensity of incident radiation.

Q.7. Define the terms (i) work function (ii) stopping potential.

Ans. (i) Work function of metal is the minimum energy required by an electron to just escape from the metal surface.

(ii) Stopping potential is that maximum negative potential given to anode in a photo-cell for which the photoelectric current becomes zero.

Q.8. What is difference between photo electric effect and thermionic emission?

Ans. In photo-electric effect, there is emission of electrons from metal surface when a light of suitable frequency falls on it. In thermionic emission, ejection of electrons from the metal surface takes place when it is suitably heated.

Q.9. What are the laws of photoelectric emission?

Ans. (i) The number of photoelectrons ejected, is directly proportional to the intensity of the incident light, i.e., if intensity of light is increased, the number of emitted electrons is increased.

(ii) The velocity of photoelectrons increases with the increase in the frequency of incident light.

(iii) The light of frequency less than a certain frequency, is incapable of ejecting electrons from metal surface. This frequency is called threshold frequency.

Q.10. What is threshold frequency?

Ans. Threshold frequency is the minimum frequency of the incident radiation for which the photo electrons are just emitted from the metal surface.

